23MAT112

Mathematics for Intelligent Systems – 2

Practice Sheet – 9

Taylor Series, Newton's Methods, Quadratic forms

- Taylor series expansion for multivariable functions in MATLAB

```
% Taylor series expansion of f(x) = sin(x)*cos(y)*exp(x) till 3rd degree of x and y
about (0,0)
syms x
syms y
taylor(sin(x)*cos(y)*exp(x),[x y],[0 0],'Order',4)
```

- Newtons method to find roots of non-linear equations:

% Code to find the root of the given non-linear system of equations and plot all the iterative points obtained. Uses separate function definition.(It can be done without defining function separately also)

$f_1(x_1, x_2) = x_1^2 + x_2^2 - 25 = 0$ and $f_2(x_1, x_2) = (x_1 - 5)^2 + (x_2 - 5)^2 - 25 = 0$

```
x=[1;2]; % starting point
plot(x(1),x(2),'*','MarkerSize', 21);% plot the starting point
hold on
for i=1:10 % 'for loop' for 10 iterations
x=x-JacobinvIntoF(x) ;
x  % to display each of the iterative points
plot(x(1),x(2),'*','MarkerSize', 21); % plot of each iterative point
pause(0.5);
hold on
axis equal
ax = gca;
ax.XAxisLocation = 'origin';
ax.YAxisLocation = 'origin';
end
hold off
function [xc]=JacobinvIntoF(x)
f1x=x(1)^2+x(2)^2-25;
f2x=(x(1)-5)^2+(x(2)-5)^2-25;
J=[2*x(1) 2*x(2);2*x(1)-10 2*x(2)-10];
xc=pinv(J)*[f1x;f2x];
end
```

- Newton's method to find the minimum of a multivariable function

  % code to find the minimum of f(x,y)=sinx+y^2+4x^2-10y+50 with starting point as (-3,5)

```
x=[-3;5]; % starting point
for i=1:5 % number of iterations
GradfAtx1x2=[cos(x(1))+8*x(1);2*x(2)-10]; % gradient at x
H=[-sin(x(1))+8,0;0,2];  % Hessian at x
xc=pinv(H)*GradfAtx1x2;
x=x-xc ;
x % to display the iterative point
Lamda=eig(H)
    % finds the eigenvalues of Hessian to check whether iterative point
    %is a minimum(positive lamdas) or a maximum(negative lamdas) or a saddle
    %point(both positive and negative lamdas)
End
```

- **Quadratic forms**: A function of the form $f(x) = x^T Ax$ is called a quadratic form, where A is a symmetric matrix. If A is positive definite the function $f(x)$ is called a convex function (a function that geometrically curves up) and if A is negative definite $f(x)$ is a concave function(a function that geometrically curves down).
  - MATLAB code for generating a positive definite matrix A and plotting the corresponding surface $f(x) = x^T Ax = 0$

```
c = 2; d = 2;
x = -10:10; y = x;
[X1, X2] = meshgrid(x,y);
% Generation of positive definite matrix using spectral decomposition, A=UDU^T.
    Eigenvalues of A are positive as diagonals of D matrix is positive.
A = [ 1, 1; 1, -1]*[c, 0; 0, d]*[1, 1; 1, -1];
Z = A(1,1)*X1.^2+ A(2,2)*X2.^2+2*A(1,2)*X1.*X2; % Z = x^T Ax
surfc(x,y,Z);
xlabel('x1'); ylabel('x2');
```

  - In a similar manner generate negative definite matrices and semidefinite matrices by changing values of 'c' and 'd' in the code and check the plot of each surface.

Practice questions

1. Using MATLAB, find the Taylor series expansion of the following functions till 3rd degree of X and Y about the given points
   (a) $f(x,y) = 3x^2 + \cos 4xy - e^{2y}$ about (2,1)
   (b) $f(x,y) = x^2 + y^2 + \frac{1}{x^2+y^2}$ at (3,5)
2. Solve the given systems of non-linear equations with the given starting points using 10 iterations of Newton's method. Also, plot them and show the exact point of intersection.
   (a) $(x-3)^2 + y^2 = 9;\ 3y - x^2 = 0, x^{(0)} = (5,6)$
   (b) $(x-3)^2 + y^2 = 9;\ 3y - x^2 = 0, x^{(0)} = (-3\,,-1)$
   (c) $(x-3)^2 + y^2 = 9;\ x^2 + (y-3)^2 = 0, x^{(0)} = (9,-3)$

3. Apply 10 iterations of Newtons method to solve the system using (1,1,1) as the starting point:
   $$-x_1^2 + x_2^2 + x_1x_2x_3 = 1.34$$
   $$x_1x_2 - x_3^2 = 0.09$$
   $$e^{x_1} - e^{x_2} + x_3 = 0.41$$
4. Apply three iterations of Newton's method to find the minimum of the given function starting from $x^{(0)} = [3, -1, 0, 1]^T$.
   $$f = (x_1 + 10x_2)^2 + 5(x_3 - x_4)^2 + (x_3 - 2x_3)^4 + 10(x_1 - x_4)^4$$
5. Write a general code for finding the minimum of a function f(x,y) using Newton's method by direct evaluation of derivatives.
6. Write a general code for finding the minimum of a function f(x,y) using Newton's method by evaluating the derivatives numerically.